

PHY110: ENGINEERING PHYSICS

FORMULA HANDBOOK & HIGH-YIELD CHEATSHEET

Complete Mathematical Formulations, Universal Constants, Derivations & Rapid
Revision Cards

Features & Exam Mastery Components

- Full Maxwell Equations, Poynting Vector, Wave Impedance & Skin Depth
 - Einstein Laser Coefficients, Population Inversion & Cavity Resonances
 - Fiber Acceptance Angle, Numerical Aperture, V-Number & Dispersion
 - Schrödinger Equations, Operators, Expectation Values & 1D Box States
- Kronig-Penney, Fermi-Dirac Distribution, Hall Coefficient & Solar Cell Parameters
 - Dielectric Polarizabilities, Clausius-Mossotti, Hysteresis & Superconductivity
 - 1-Page Rapid Recall Cheatsheet per Unit for Under 10-Minute Revisions

Department of Physics & Electrical Sciences

Lovely Professional University, Phagwara, Punjab

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Contents

1	Fundamental Physical Constants & Unit Conversions	1
2	Unit 1: Electromagnetic Theory & Vector Calculus	3
3	Unit 2: Lasers & Optical Resonance	6
4	Unit 3: Fiber Optics & Waveguides	8
5	Unit 4: Quantum Mechanics	10
6	Unit 5: Semiconductor Physics & Solid State	12
7	Unit 6: Engineering Materials & Superconductivity	14
8	Rapid Exam Revision: 1-Page Summary Cheatsheet	16

Chapter 1

Fundamental Physical Constants & Unit Conversions

Table of Universal Physical Constants

Constant Name	Symbol	Value in SI Units	Alternative / Common Units
Speed of Light in Vacuum	c	$2.99792458 \times 10^8 \text{ m/s} \approx 3 \times 10^8 \text{ m/s}$	$3 \times 10^{10} \text{ cm/s}$
Planck Constant	h	$6.62607015 \times 10^{-34} \text{ J} \cdot \text{s}$	$4.135667 \times 10^{-15} \text{ eV} \cdot \text{s}$
Reduced Planck Constant	$\hbar = h/(2\pi)$	$1.0545718 \times 10^{-34} \text{ J} \cdot \text{s}$	$6.582119 \times 10^{-16} \text{ eV} \cdot \text{s}$
Elementary Electron Charge	e	$1.602176634 \times 10^{-19} \text{ C}$	—
Electron Rest Mass	m_e	$9.1093837 \times 10^{-31} \text{ kg}$	$0.5109989 \text{ MeV}/c^2$
Proton Rest Mass	m_p	$1.6726219 \times 10^{-27} \text{ kg}$	$938.272 \text{ MeV}/c^2$
Neutron Rest Mass	m_n	$1.6749275 \times 10^{-27} \text{ kg}$	$939.565 \text{ MeV}/c^2$
Permittivity of Free Space	ϵ_0	$8.8541878 \times 10^{-12} \text{ F/m}$	$\frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ H/m} \approx 1.2566 \times 10^{-6} \text{ N/A}^2$	$\mu_0\epsilon_0 = 1/c^2$
Free Space Wave Impedance	$\eta_0 = \sqrt{\mu_0/\epsilon_0}$	$376.73 \Omega \approx 120\pi \Omega \approx 377 \Omega$	—
Boltzmann Constant	k_B	$1.380649 \times 10^{-23} \text{ J/K}$	$8.617333 \times 10^{-5} \text{ eV/K}$
Avogadro Number	N_A	$6.02214076 \times 10^{23} \text{ mol}^{-1}$	—
Bohr Magneton	$\mu_B = \frac{e\hbar}{2m_e}$	$9.274009 \times 10^{-24} \text{ J/T}$	$5.788381 \times 10^{-5} \text{ eV/T}$
Bohr Radius	$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$	$5.291772 \times 10^{-11} \text{ m} = 0.529 \text{ Å}$	—
Rydberg Constant	R_∞	$1.097373 \times 10^7 \text{ m}^{-1}$	13.60569 eV

Crucial Energy and Dimensional Conversion Formulas

- **Electron-Volt to Joules:** $1 \text{ eV} = 1.60218 \times 10^{-19} \text{ J} \iff 1 \text{ J} = 6.2415 \times 10^{18} \text{ eV}$.
- **Photon Energy vs Wavelength:** $E = h\nu = \frac{hc}{\lambda} = \frac{1240 \text{ eV}\cdot\text{nm}}{\lambda \text{ (in nm)}} = \frac{12400 \text{ eV}\cdot\text{\AA}}{\lambda \text{ (in \AA)}}$.
- **Thermal Energy at Room Temperature ($T = 300 \text{ K}$):** $k_B T \approx 0.02585 \text{ eV} \approx 26 \text{ meV} \approx 4.14 \times 10^{-21} \text{ J}$.
- **de Broglie Wavelength of Electron accelerated through V Volts:**

$$\lambda_e = \frac{h}{\sqrt{2m_e eV}} = \frac{1.227}{\sqrt{V}} \text{ nm} = \sqrt{\frac{150}{V}} \text{ \AA}$$

Chapter 2

Unit 1: Electromagnetic Theory & Vector Calculus

1. Vector Differential Operators in Cartesian Coordinates

- **Del Operator:** $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$
- **Gradient (Scalar field ϕ):** $\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$
- **Divergence (Vector field $\vec{A}$):** $\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$
- **Curl (Vector field $\vec{A}$):** $\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$
- **Laplacian Operator:** $\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$
- **Vector Curl of Curl Identity:** $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$
- **Zero Identities:** $\nabla \times (\nabla \phi) = \vec{0}$ (conservative fields), $\nabla \cdot (\nabla \times \vec{A}) = 0$ (solenoidal fields).

2. Maxwell's Equations (Differential & Integral Forms)

Law / Principle	Differential Form	Integral Form	Physical Meaning
Gauss's Law (Electrostatics)	$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{A} = Q_{\text{enc}}$	Electric charges are flux sources/sinks.
Gauss's Law (Magnetism)	$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{A} = 0$	No isolated magnetic monopoles exist.
Faraday's Law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$	Time-varying $\vec{B}$ induces circulating $\vec{E}$.
Ampere-Maxwell Law	$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t}$	$\oint_S \vec{H} \cdot d\vec{\ell} = I_{\text{enc}} + \int_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{A}$	Conduction + displacement current generate $\vec{B}$.

3. Charge Continuity, Displacement Current & Poynting Vector

- Equation of Continuity:** $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \implies \oint_S \vec{J} \cdot d\vec{A} = -\frac{dQ}{dt}$
- Displacement Current Density:** $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon \frac{\partial \vec{E}}{\partial t}$
- Total Current across Capacitor Plates:** $I_d = \epsilon_0 A \frac{dE}{dt} = C \frac{dV}{dt} = I_c$
- Poynting Vector (Instantaneous Energy Flux):** $\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$ [W/m²]
- Time-Average Poynting Flux:** $\langle S \rangle = \frac{1}{2} \text{Re}(\vec{E} \times \vec{H}^*) = \frac{E_0^2}{2\eta} = \frac{1}{2} \epsilon v E_0^2$
- Poynting Theorem (Energy Conservation):** $-\oint_S \vec{S} \cdot d\vec{A} = \int_V \vec{J} \cdot \vec{E} dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV$

4. Electromagnetic Wave Propagation in Linear Media

- Wave Equation (Lossless):** $\nabla^2 \vec{E} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0$, with phase velocity $v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{n}$
- Intrinsic Wave Impedance:** $\eta = \sqrt{\frac{\mu}{\epsilon}}$, for free space $\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377 \Omega$
- Lossy Media Complex Wave Number:** $\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$
- Good Conductor Limit ($\sigma \gg \omega\epsilon$):**

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\pi f \mu \sigma}, \quad \text{Skin Depth } \delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

- Radiation Pressure:** $P = \frac{\langle S \rangle}{c}$ (perfect absorption), $P = \frac{2\langle S \rangle}{c}$ (perfect reflection).

Quick Exam Trap: Conduction current $\vec{J}_c = \sigma \vec{E}$ and displacement current $\vec{J}_d = j\omega\epsilon\vec{E}$ have a phase difference of exactly 90° ($\pi/2$). The loss tangent is $\tan \delta = \frac{|\vec{J}_c|}{|\vec{J}_d|} = \frac{\sigma}{\omega\epsilon}$.

Chapter 3

Unit 2: Lasers & Optical Resonance

1. Einstein's Phenomenological Coefficients & Relations

- **Transition Rates:**

$$\text{Stimulated Absorption Rate: } R_{12} = B_{12}N_1u(\nu)$$

$$\text{Spontaneous Emission Rate: } R_{21}^{\text{spont}} = A_{21}N_2$$

$$\text{Stimulated Emission Rate: } R_{21}^{\text{stim}} = B_{21}N_2u(\nu)$$

- **Einstein Relations (Thermodynamic Equilibrium with Planck's Law):**

$$B_{12} = \frac{g_2}{g_1} B_{21} \quad (\text{if } g_1 = g_2, B_{12} = B_{21})$$

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} = \frac{8\pi h}{\lambda^3}$$

- **Ratio of Spontaneous to Stimulated Emission Rate:**

$$\frac{R_{\text{spont}}}{R_{\text{stim}}} = \frac{A_{21}}{B_{21}u(\nu)} = e^{h\nu/k_B T} - 1$$

At optical frequencies ($\nu \sim 5 \times 10^{14}$ Hz) and room temp, $h\nu \gg k_B T \implies$ Spontaneous emission dominates without an active pumping cavity.

2. Population Inversion, Threshold Condition & Gain Coefficient

- **Boltzmann Population Distribution:** $\frac{N_2}{N_1} = e^{-(E_2-E_1)/k_B T} < 1$ (for $E_2 > E_1, T > 0$).

- **Population Inversion Condition:** $N_2 > \frac{g_2}{g_1} N_1$.

- **Small-Signal Gain Coefficient:** $\gamma(\nu) = \sigma_{21}(\nu) \left[N_2 - \frac{g_2}{g_1} N_1 \right]$.

- **Optical Intensity Amplification along Medium:** $I(z) = I_0 e^{[\gamma(\nu) - \alpha_s]z}$.

- **Threshold Condition for Sustained Oscillation in Cavity (Length L , mirror reflectivities R_1, R_2):**

$$\gamma_{\text{th}} = \alpha_s + \frac{1}{2L} \ln \left(\frac{1}{R_1 R_2} \right)$$

- **Fabry-Perot Longitudinal Cavity Modes:** $L = m \frac{\lambda_m}{2} \implies \nu_m = m \frac{c}{2nL}, \quad \Delta\nu = \nu_{m+1} - \nu_m = \frac{c}{2nL}$.

3. Comparison of Common Laser Systems

Laser Type	Active Medium	Pumping	Wavelength	Key Characteristics
Ruby Laser	$\text{Cr}^{3+}:\text{Al}_2\text{O}_3$	Optical (Xenon Flash)	694.3 nm (Red)	3-level pulsed system, high peak power.
He-Ne Laser	He : Ne (10:1 gas)	Electric Discharge	632.8 nm (Red)	4-level CW laser, high spatial coherence.
Nd:YAG	$\text{Nd}^{3+}:\text{Y}_3\text{Al}_5\text{O}_{12}$	Optical (Krypton/LED)	1064 nm (NIR)	4-level high efficiency CW/pulsed system.
Semiconductor	GaAs / InP junction	Direct Current Injection	840–1550 nm	Ultra-compact, high electrical-to-optical conversion.

4. Holography (Gabor Principle)

- **Recording:** Interference of coherent Reference beam $R = A_r e^{i\phi_r}$ and Object beam $O = A_o e^{i\phi_o}$:

$$I(x, y) = |R + O|^2 = |R|^2 + |O|^2 + R^*O + RO^*$$

- **Reconstruction:** Illuminating recorded hologram transmittance $t \propto I(x, y)$ with original reference beam R :

$$R \cdot I(x, y) = R(|R|^2 + |O|^2) + |R|^2 O + R^2 O^*$$

The term $|R|^2 O$ generates the **virtual 3D image** (exact true object wave reconstructed in space).

Chapter 4

Unit 3: Fiber Optics & Waveguides

1. Total Internal Reflection, Acceptance Angle & Numerical Aperture

- **Critical Angle at Core-Cladding Interface:** $\sin \theta_c = \frac{n_2}{n_1}$ ($n_1 > n_2$).
- **Maximum Acceptance Angle (θ_a in medium of index n_0):**

$$\sin \theta_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

For air launcher ($n_0 = 1$): $\theta_a = \arcsin \left(\sqrt{n_1^2 - n_2^2} \right)$.

- **Numerical Aperture (NA):**

$$\text{NA} = \sin \theta_a = \sqrt{n_1^2 - n_2^2} \approx n_1 \sqrt{2\Delta}, \quad \text{where } \Delta = \frac{n_1 - n_2}{n_1} \approx \frac{n_1^2 - n_2^2}{2n_1^2}$$

Δ is the fractional refractive index change (≈ 0.005 – 0.01).

2. V-Number (Normalized Frequency) & Modal Capacity

- **V-Number Definition (Core radius a , diameter $d = 2a$):**

$$V = \frac{2\pi a}{\lambda_0} \text{NA} = \frac{\pi d}{\lambda_0} \sqrt{n_1^2 - n_2^2}$$

- **Single-Mode Cutoff Condition (Step-Index):** $V < 2.405$.
- **Number of Bound Guided Modes (M):**

$$\begin{aligned} \text{Step-Index Fiber (SIF): } M_{\text{SIF}} &\approx \frac{V^2}{2} \\ \text{Graded-Index Fiber (GRIN, Parabolic } \alpha = 2\text{): } M_{\text{GRIN}} &\approx \frac{V^2}{4} = \frac{1}{2} M_{\text{SIF}} \end{aligned}$$

3. Dispersion & Pulse Broadening

- **Intermodal Dispersion in Step-Index Fiber (Time delay per length):**

$$\Delta\tau_{\text{modal}} = \frac{L}{c} \left(\frac{n_1^2 - n_2 n_1}{n_2} \right) \approx \frac{L n_1 \Delta}{c} = \frac{L (\text{NA})^2}{2 n_1 c} \quad [\text{s}]$$

- **Intermodal Dispersion in Graded-Index Fiber (Optimal $\alpha = 2 - 2\Delta$):**

$$\Delta\tau_{\text{GRIN}} \approx \frac{Ln_1\Delta^2}{8c} \implies \text{Reduced by factor of } \frac{\Delta}{8} \approx 100\text{--}1000\times$$

- **Material Dispersion (Wavelength dependent $n(\lambda)$):** $\Delta\tau_{\text{mat}} = -\frac{L\lambda}{c} \left(\frac{d^2n}{d\lambda^2} \right) \Delta\lambda$.

4. Attenuation & Optical Transmission Windows

- **Attenuation Coefficient (α in dB/km):**

$$\alpha_{\text{dB/km}} = \frac{10}{L \text{ (km)}} \log_{10} \left(\frac{P_{\text{in}}}{P_{\text{out}}} \right) \iff P_{\text{out}} = P_{\text{in}} \cdot 10^{-\alpha L/10}$$

- **Rayleigh Scattering Loss:** $\alpha_{\text{Rayleigh}} \propto \frac{1}{\lambda^4}$.
- **Standard Optical Communication Windows:**
 1. **1st Window (850 nm):** $\alpha \approx 2\text{--}3$ dB/km (GaAs emitters, multimode LANs).
 2. **2nd Window (1310 nm):** $\alpha \approx 0.35\text{--}0.5$ dB/km (Zero material dispersion in silica).
 3. **3rd Window (1550 nm):** $\alpha \approx 0.18\text{--}0.20$ dB/km (Global minimum attenuation, EDFA amplification band).

Chapter 5

Unit 4: Quantum Mechanics

1. Wave-Particle Duality & de Broglie Hypothesis

- **Planck-Einstein Relation:** $E = h\nu = \hbar\omega$.
- **de Broglie Wavelength:** $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE_k}}$.
- **Thermal Neutron / Gas Particle Wavelength:** $\lambda_{\text{th}} = \frac{h}{\sqrt{3mk_B T}}$.
- **Phase Velocity vs Group Velocity:**

$$v_p = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk} = v_{\text{particle}} \implies v_p \cdot v_g = c^2 \quad (\text{for relativistic matter waves})$$

- **Bragg's Diffraction Law:** $2d \sin \theta = n\lambda$.

2. Heisenberg Uncertainty Principle

- **Position-Momentum Uncertainty:** $\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$.
- **Energy-Time Uncertainty:** $\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$ (determines atomic transition natural linewidth $\Delta\nu = \frac{1}{2\pi\Delta t}$).
- **Angular Position-Angular Momentum:** $\Delta\theta \cdot \Delta L_z \geq \frac{\hbar}{2}$.

3. Schrödinger Wave Equations & Quantum Operators

- **Quantum Mechanical Operators in Coordinate Representation:**

$$\text{Position Operator: } \hat{x} = x$$

$$\text{Momentum Operator: } \hat{p}_x = -i\hbar \frac{\partial}{\partial x} \implies \hat{\vec{p}} = -i\hbar \nabla$$

$$\text{Energy / Hamiltonian: } \hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}), \quad \hat{E} = i\hbar \frac{\partial}{\partial t}$$

- **Time-Dependent Schrödinger Equation (TDSE):**

$$i\hbar \frac{\partial \Psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\vec{r}, t) + V(\vec{r}) \Psi(\vec{r}, t)$$

- **Time-Independent Schrödinger Equation (TISE):**

$$\nabla^2 \psi(\vec{r}) + \frac{2m}{\hbar^2} [E - V(\vec{r})] \psi(\vec{r}) = 0$$

- **Born Statistical Interpretation & Normalization:**

$$P(x) dx = |\psi(x)|^2 dx, \quad \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

- **Expectation Value:** $\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) dx$.

4. Particle in a 1D Infinite Potential Well ($V(x) = 0$ for $0 \leq x \leq L$)

- **Normalized Wavefunctions:** $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \dots$

- **Energy Eigenvalues:**

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2} \implies E_n = n^2 E_1$$

- **Ground State (Zero-Point Energy):** $E_1 = \frac{h^2}{8mL^2} > 0$ (direct consequence of Heisenberg uncertainty).

Chapter 6

Unit 5: Semiconductor Physics & Solid State

1. Classical Drude-Lorentz Free Electron Theory

- **Drift Velocity & Mobility:** $v_d = \mu E = \frac{e\tau}{m} E \implies \mu = \frac{e\tau}{m}$.
- **Electrical Conductivity:** $\sigma = ne\mu = \frac{ne^2\tau}{m}$.
- **Wiedemann-Franz Law & Lorentz Number:**

$$\frac{K}{\sigma T} = L = \frac{\pi^2 k_B^2}{3e^2} \approx 2.44 \times 10^{-8} \text{ W} \cdot \Omega / \text{K}^2$$

2. Band Theory of Solids (Kronig-Penney Model)

- **Bloch Theorem:** $\psi_k(x) = e^{ikx} u_k(x)$, where $u_k(x+a) = u_k(x)$.
- **Kronig-Penney Dispersion Relation:**

$$P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos(ka), \quad \text{where } P = \frac{mV_0ba}{\hbar^2}, \alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

- If $P \rightarrow 0$: Free electron parabola $E = \frac{\hbar^2 k^2}{2m}$.
- If $P \rightarrow \infty$: Discrete atomic levels $\alpha a = n\pi \implies E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$.
- **Effective Mass of Electron:** $m^* = \frac{\hbar^2}{d^2 E / dk^2}$.

3. Fermi-Dirac Distribution & Carrier Statistics

- **Fermi-Dirac Occupation Probability:** $f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$.
- **Intrinsic Carrier Concentration:** $n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$.
- **Intrinsic Fermi Level:** $E_F = \frac{E_c + E_v}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_h^*}{m_e^*}\right) \approx \text{Midgap}$.
- **Extrinsic Fermi Levels:**

$$\text{N-type: } E_F = E_c - k_B T \ln\left(\frac{N_c}{N_d}\right), \quad \text{P-type: } E_F = E_v + k_B T \ln\left(\frac{N_v}{N_a}\right)$$

- **Mass Action Law (Equilibrium):** $n \cdot p = n_i^2 = N_c N_v e^{-E_g/k_B T}$.

4. Hall Effect & Solar Cells

- **Hall Voltage:** $V_H = \frac{BI}{nqw} = \frac{R_H BI}{w}$.
- **Hall Coefficient:** $R_H = \frac{1}{nq} = \begin{cases} -\frac{1}{ne} & \text{(N-type electrons)} \\ +\frac{1}{pe} & \text{(P-type holes)} \end{cases}$
- **Hall Mobility:** $\mu_H = |R_H|\sigma$.
- **Solar Cell Fill Factor & Conversion Efficiency:**

$$FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}, \quad \eta = \frac{P_{max}}{P_{in}} = \frac{FF \cdot V_{oc} \cdot I_{sc}}{P_{in} \cdot A_{cell}}$$

Chapter 7

Unit 6: Engineering Materials & Superconductivity

1. Dielectric Materials & Polarization

- **Electric Displacement Vector:** $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_r \epsilon_0 \vec{E}$.
- **Electric Susceptibility:** $\vec{P} = \epsilon_0 \chi_e \vec{E} \implies \epsilon_r = 1 + \chi_e$.
- **Lorentz Local Internal Field (Cubic Lattice):** $E_{\text{loc}} = E + \frac{P}{3\epsilon_0}$.
- **Total Atomic Polarizability:** $\alpha = \alpha_e + \alpha_i + \alpha_o + \alpha_s$.

Electronic Polarizability: $\alpha_e = 4\pi\epsilon_0 R^3$ (Temperature-independent)

Orientational Polarizability (Langevin): $\alpha_o = \frac{p_0^2}{3k_B T}$ ($\propto 1/T$)

- **Clausius-Mossotti Equation:**

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} = \frac{N(\alpha_e + \alpha_i)}{3\epsilon_0}$$

- **Dielectric Loss Tangent:** $\tan \delta = \frac{\epsilon''}{\epsilon'}$.

2. Magnetic Materials & Hysteresis

- **Magnetic Flux Density:** $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_r \mu_0 \vec{H}$.
- **Magnetic Susceptibility:** $\vec{M} = \chi_m \vec{H} \implies \mu_r = 1 + \chi_m$.
- **Curie-Weiss Law (Ferromagnet $\rightarrow$ Paramagnet above T_C):** $\chi_m = \frac{C}{T - \theta_p}$.
- **Hysteresis Energy Loss per Cycle:** $W_h = \oint \vec{B} \cdot d\vec{H} = \text{Area of } B\text{-}H \text{ Loop}$.

3. Superconductivity & Meissner Effect

- **Critical Magnetic Field Parabolic Temperature Law:**

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

- **Silsbee's Critical Current Rule (Wire radius r):** $I_c = 2\pi r H_c$.

- **Meissner Effect (Perfect Diamagnetism):** $\vec{B} = 0 \implies \mu_0(\vec{H} + \vec{M}) = 0 \implies \vec{M} = -\vec{H} \implies \chi_m = -1$.
- **London Penetration Depth:** $\lambda_L = \sqrt{\frac{m}{\mu_0 n_s e^2}}$, $B(x) = B(0)e^{-x/\lambda_L}$.
- **BCS Superconducting Energy Gap at 0 K:** $E_g(0) = 2\Delta(0) = 3.528 k_B T_c$.
- **Josephson Supercurrent:** $I_s = I_c \sin(\Delta\phi)$, with AC frequency $f = \frac{2eV}{h} = \frac{V}{\Phi_0}$ ($\Phi_0 = \frac{h}{2e} \approx 2.067 \times 10^{-15}$ Wb).

Chapter 8

Rapid Exam Revision: 1-Page Summary Cheatsheet

Rapid Formula Matrix for PHY110

Topic / Area	Key Master Equation	Exam Tip / Core Takeaway
Maxwell 4	$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t}$	Maxwell added $\frac{\partial \vec{D}}{\partial t}$ to satisfy continuity $\nabla \cdot \vec{J} + \dot{\rho} = 0$.
Poynting Flux	$\vec{S} = \vec{E} \times \vec{H}, \quad \langle S \rangle = \frac{E_0^2}{2\eta_0}$	$\eta_0 = \sqrt{\mu_0/\epsilon_0} = 377 \Omega$. Poynting is in W/m^2 .
Skin Depth	$\delta = \sqrt{2/(\omega\mu\sigma)} = 1/\sqrt{\pi f \mu \sigma}$	High frequency RF current flows only on the outer conductor sheath.
Einstein Laser	$\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3}, \quad B_{12} = B_{21}$	A_{21} is spontaneous emission; B_{21} is stimulated emission.
Laser Cavity	$\gamma_{\text{th}} = \alpha_s + \frac{1}{2L} \ln\left(\frac{1}{R_1 R_2}\right)$	Population inversion $N_2 > N_1$ requires negative temperature analog.
He-Ne Laser	$\lambda = 632.8 \text{ nm (Red)}, \text{ He:Ne} = 10:1$	He absorbs discharge energy and transfers to Ne via resonant collision.
Numerical Aperture	$\text{NA} = \sin \theta_a = \sqrt{n_1^2 - n_2^2} = n_1 \sqrt{2\Delta}$	Measures fiber light collection capability. Independent of fiber diameter!
V-Number	$V = \frac{\pi d}{\lambda} \text{NA}, \quad V < 2.405 \text{ for Single Mode}$	Total modes in step-index $M \approx V^2/2$, in GRIN $M \approx V^2/4$.
Optical Windows	850 nm, 1310 nm, 1550 nm	Lowest loss at 1550 nm (0.2 dB/km); Zero dispersion at 1310 nm.
de Broglie	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{1.227}{\sqrt{V}} \text{ nm}$	Verified by Davisson-Germer electron nickel crystal scattering.
1D Box Energy	$E_n = \frac{n^2 h^2}{8mL^2}, \quad \psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$	Ground state energy $E_1 > 0$ due to Heisenberg uncertainty $\Delta x \Delta p \geq \hbar/2$.
Fermi-Dirac	$f(E) = \frac{1}{1 + e^{(E - E_F)/k_B T}}$	At $T = 0 \text{ K}$, $f(E < E_F) = 1$ and $f(E > E_F) = 0$. At $E = E_F$, $f = 0.5$.
Hall Effect	$R_H = \frac{1}{nq} = \frac{V_H w}{BI}$	$R_H < 0 \implies$ N-type (electrons); $R_H > 0 \implies$ P-type (holes).
Mass Action	$n \cdot p = n_i^2 = N_c N_v e^{-E_g/k_B T}$	Holds for both intrinsic and extrinsic semiconductors in thermal equilibrium.
Clausius-Mossotti	$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$	Connects microscopic polarizability α with macroscopic dielectric constant ϵ_r .
Meissner Effect	$\vec{B} = 0 \implies \chi_m = -1$	Superconductors are ideal diamagnets below T_c .
Critical Field	$H_c(T) = H_c(0)[1 - (T/T_c)^2]$	Parabolic decrease of critical field with increasing temperature.